

SI 140A: Probability and Statistics for EECS

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Midterm, 2025/04/15

Midterm Exam

Time Limit: 100 Minutes

Name: _____

Student ID: _____

This exam contains 9 pages (including this cover page) and 8 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

Try to answer as many problems as you can. The following rules apply:

- All examination rules must be strictly observed throughout the entire test, and any form of cheating is prohibited.
- Other than allowable materials, students taking closed-book tests must place their books, notes, tablets, and any other electronic devices in places designated by the examiners.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.

Do not write in the table below.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	20	
8	20	
Total:	100	

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Problem 1

- (a) (5') How many ways are there to distribute 15 distinct books into 3 groups, where one group has 3 books and the other two groups have 6 books each?

Notice: The groups with same sizes here are undistinguished here. You may write the answers in a form with combinatorial numbers.

- (b) (5') How many distinct **positive** integer-valued vectors (x_1, x_2, x_3, x_4) satisfy the following equation and inequalities?

$$x_1 + x_2 + x_3 + x_4 = 40$$

$$x_1 > 5, x_2 > 4, x_3 > 2, x_4 > 3$$

Problem 2

(10') A family has two children. We assume that each child has black hair with probability p , independent of each other and of gender. What is the probability that both children are girls given that at least one is a black-hair girl?

Problem 3

(10') Let $X_1, X_2, \dots$ be independent $\mathcal{N}(0, 4)$ r.v.s., and let J be the smallest value of j such that $X_j > 4$ (i.e. the index of the first X_j exceeding 4). In terms of Φ (the CDF of the **standard** normal variable), find $E[J]$.

Problem 4

(10') Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X .

Problem 5

Let X be a random variable (discrete or continuous) such that $0 \leq X \leq 1$ always holds. Let $\mu = E[X]$.

- (a) (5') Show that

$$\text{Var}[X] \leq \mu - \mu^2 \leq \frac{1}{4}.$$

(*Hint: With probability 1, we have $X^2 \leq X$.*)

- (b) (5') Show that there is only one possible distribution for X for which $\text{Var}[X] = \frac{1}{4}$. What is the name of this distribution?

Problem 6

(10') In a movie theater, there is a line (queue) to buy the ticket. A manager announces that she will give a free ticket to the first person in line whose birthday is the same as someone who has already bought a ticket. Now, since you are a VIP of this theater and have the opportunity to choose any position in the line. Assuming that you do not know anyone else's birthday and all birthdays are randomly distributed throughout the year (365 days in a year), which position in line gives you the best chance of getting the free ticket (i.e. the highest probability)?

Problem 7

Suppose a biased coin is tossed repeatedly with a probability of getting a head as p , and we obtain a sequence of H and T (H denotes Head and T denotes Tail). Let N denote the number of tosses to observe the first occurrence of the pattern “HHH”.

- (a) (6') Find the recurrence relation of $P(N = n)$.
- (b) (7') Given $p = \frac{1}{3}$, find $E[N]$.
- (c) (7') Given $p = \frac{1}{3}$, find $Var[N]$.

Problem 8

Let $Z \sim \mathcal{N}(0, 1)$. A measuring device is used to observe Z , but the device can only handle positive values and gives a reading of 0 if $Z \leq 0$. This is an example of censored data. So assume that

$$X = Z \cdot I\{Z \geq 0\}$$

is observed rather than Z , where $I\{Z > 0\}$ is the indicator of the event $Z > 0$.

- (a) (6') Find $E[X]$. Note that X is neither continuous nor discrete but some rules still apply.
- (b) (7') Find $Var(X)$.
- (c) (7') We can write X in the form $X = \max(Z, 0)$. Generally, for $Y = \max(Z - c, 0)$, where $c \geq 0$, find $E[Y]$ in terms of the standard normal PDF ϕ and its CDF Φ .

Y discrete		Y continuous	
X discrete	$P(Y = y X = x) = \frac{P(X=x Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y X = x) = \frac{P(X=x Y=y)f_Y(y)}{P(X=x)}$	
X continuous	$P(Y = y X = x) = \frac{f_X(x Y=y)P(Y=y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$	

Y discrete		Y continuous	
X discrete	$P(X = x) = \sum_y P(X = x Y = y)P(Y = y)$	$P(X = x) = \int_{-\infty}^{\infty} P(X = x Y = y)f_Y(y)dy$	
X continuous	$f_X(x) = \sum_y f_X(x Y = y)P(Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X Y}(x y)f_Y(y)dy$	

Figure 1: The Bayes' Rule and LOTP.

Name	Param.	PMF or PDF	Mean	Variance
Bernoulli	p	$P(X = 1) = p, P(X = 0) = q$	p	pq
Binomial	n, p	$\binom{n}{k}p^kq^{n-k}$, for $k \in \{0, 1, \dots, n\}$	np	npq
FS	p	pq^{k-1} , for $k \in \{1, 2, \dots\}$	$1/p$	q/p^2
Geom	p	pq^k , for $k \in \{0, 1, 2, \dots\}$	q/p	q/p^2
NBinom	r, p	$\binom{r+n-1}{r-1}p^r q^n$, $n \in \{0, 1, 2, \dots\}$	rq/p	rq/p^2
HGeom	w, b, n	$\frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}$, for $k \in \{0, 1, \dots, n\}$	$\mu = \frac{nw}{w+b}$	$\left(\frac{w+b-n}{w+b-1}\right)n\frac{\mu}{n}(1 - \frac{\mu}{n})$
Poisson	λ	$\frac{e^{-\lambda}\lambda^k}{k!}$, for $k \in \{0, 1, 2, \dots\}$	λ	λ
Uniform	$a < b$	$\frac{1}{b-a}$, for $x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	μ, σ^2	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2
Log-Normal	μ, σ^2	$\frac{1}{x\sigma\sqrt{2\pi}}e^{-(\log x - \mu)^2/(2\sigma^2)}$, $x > 0$	$\theta = e^{\mu+\sigma^2/2}$	$\theta^2(e^{\sigma^2} - 1)$
Expo	λ	$\lambda e^{-\lambda x}$, for $x > 0$	$1/\lambda$	$1/\lambda^2$
Gamma	a, λ	$\Gamma(a)^{-1}(\lambda x)^a e^{-\lambda x} x^{a-1}$, for $x > 0$	a/λ	a/λ^2
Beta	a, b	$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}x^{a-1}(1-x)^{b-1}$, for $0 < x < 1$	$\mu = \frac{a}{a+b}$	$\frac{\mu(1-\mu)}{a+b+1}$
Chi-Square	n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$, for $x > 0$	n	$2n$
Student- t	n	$\frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)}(1 + x^2/n)^{-(n+1)/2}$	0 if $n > 1$	$\frac{n}{n-2}$ if $n > 2$

Figure 2: Some properties of distributions.